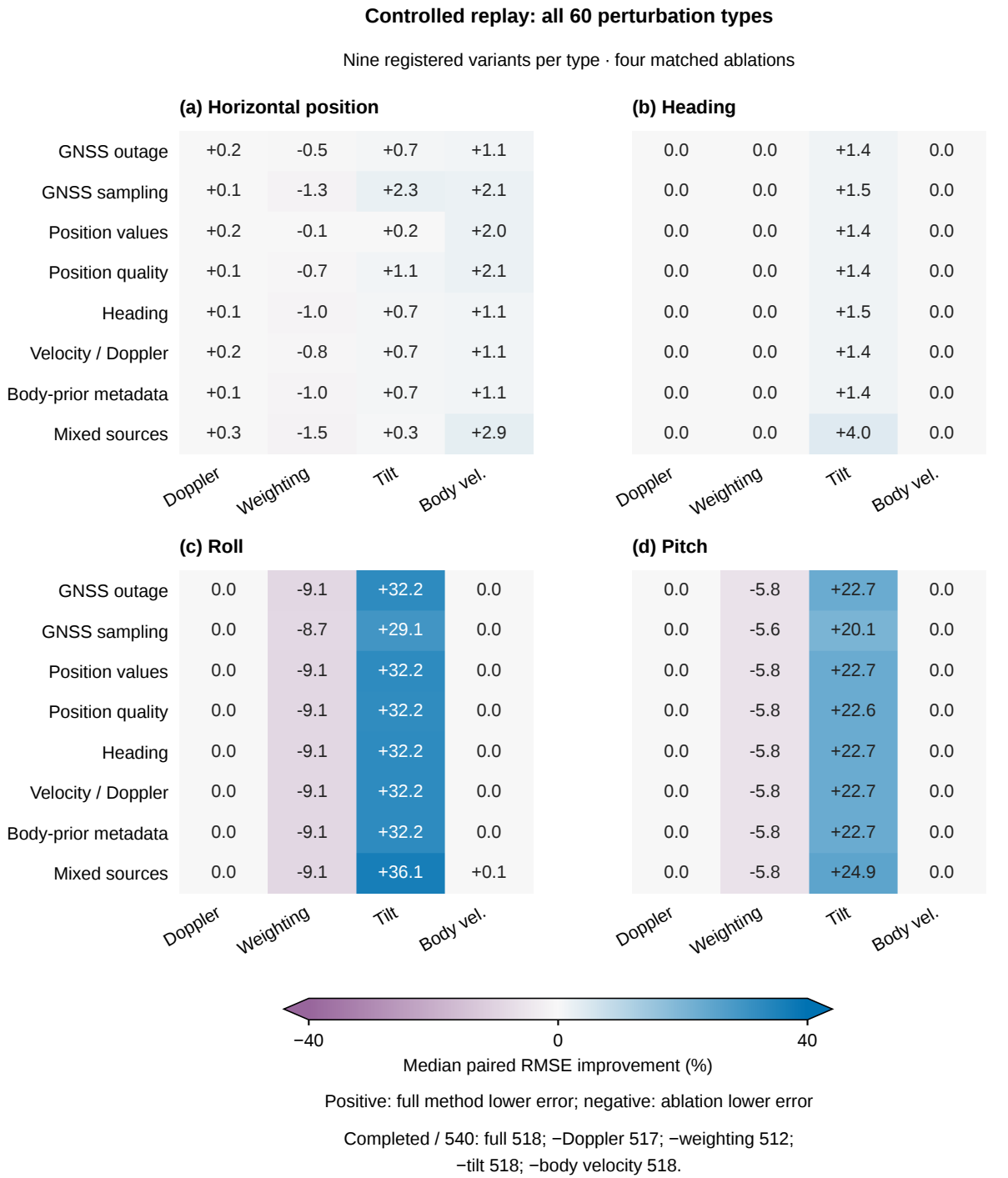
Controlled replay: all 60 perturbation types
Nine registered variants per type · four matched ablations
(a) Horizontal position
GNSS outage +0.2 -0.5 +0.7 +1.1
GNSS sampling +0.1 -1.3 +2.3 +2.1
Position values +0.2 -0.1 +0.2 +2.0
Position quality +0.1 -0.7 +1.1 +2.1
Heading +0.1 -1.0 +0.7 +1.1
Velocity / Doppler +0.2 -0.8 +0.7 +1.1
Body-prior metadata +0.1 -1.0 +0.7 +1.1
Mixed sources +0.3 -1.5 +0.3 +2.9
Doppler
Weighting
Tilt
Body vel.
(b) Heading
0.0 0.0 +1.4 0.0
0.0 0.0 +1.5 0.0
0.0 0.0 +1.4 0.0
0.0 0.0 +1.4 0.0
0.0 0.0 +1.5 0.0
0.0 0.0 +1.4 0.0
0.0 0.0 +1.4 0.0
0.0 0.0 +4.0 0.0
Doppler
Weighting
Tilt
Body vel.
(c) Roll
GNSS outage 0.0 -9.1 +32.2 0.0
GNSS sampling 0.0 -8.7 +29.1 0.0
Position values 0.0 -9.1 +32.2 0.0
Position quality 0.0 -9.1 +32.2 0.0
Heading 0.0 -9.1 +32.2 0.0
Velocity / Doppler 0.0 -9.1 +32.2 0.0
Body-prior metadata 0.0 -9.1 +32.2 0.0
Mixed sources 0.0 -9.1 +36.1 +0.1
Doppler
Weighting
Tilt
Body vel.
(d) Pitch
0.0 -5.8 +22.7 0.0
0.0 -5.6 +20.1 0.0
0.0 -5.8 +22.7 0.0
0.0 -5.8 +22.6 0.0
0.0 -5.8 +22.7 0.0
0.0 -5.8 +22.7 0.0
0.0 -5.8 +22.7 0.0
0.0 -5.8 +24.9 0.0
Doppler
Weighting
Tilt
Body vel.
−40
0
40
Median paired RMSE improvement (%)
Positive: full method lower error; negative: ablation lower error
Completed / 540: full 518; −Doppler 517; −weighting 512;
−tilt 518; −body velocity 518.